

```
#Load needed libraries
```

```
library(ggplot2)
```

```
library(tidyverse)
```

```
# Problem 1: Generate a list of datasets within R itself
```

```
> ls("package:datasets")
```

```
[1] "ability.cov"      "airmiles"         "AirPassengers"    "airquality"       "anscombe"
[6] "attenu"           "attitude"        "austres"           "beaver1"           "beaver2"
[11] "BJsales"          "BJsales.lead"     "BOD"               "cars"              "ChickWeight"
[16] "chickwts"         "co2"              "CO2"               "crimtab"           "discoveries"
[21] "DNase"            "esoph"            "euro"              "euro.cross"        "eurodist"
[26] "EuStockMarkets"   "faithful"         "fdeaths"           "Formaldehyde"     "freeny"
[31] "freeny.x"         "freeny.y"         "gait"              "HairEyeColor"      "Harman23.cor"
[36] "Harman74.cor"     "Indometh"         "infert"            "InsectSprays"      "iris"
[41] "iris3"            "islands"          "JohnsonJohnson"  "LakeHuron"         "ldeaths"
[46] "lh"               "LifeCycleSavings" "Loblolly"          "longley"           "lynx"
[51] "mdeaths"         "morley"           "mtcars"            "nhtemp"            "Nile"
[56] "nottem"           "npk"              "occupationalStatus" "Orange"            "OrchardSprays"
[61] "penguins"        "penguins_raw"     "PlantGrowth"       "precip"            "presidents"
[66] "pressure"        "Puromycin"        "quakes"            "randu"             "rivers"
[71] "rock"            "Seatbelts"        "sleep"             "stack.loss"        "stack.x"
[76] "stackloss"       "state.abb"        "state.area"        "state.center"      "state.division"
[81] "state.name"      "state.region"     "state.x77"         "sunspot.m2014"     "sunspot.month"
[86] "sunspot.year"    "sunspots"         "swiss"             "Theoph"            "Titanic"
[91] "ToothGrowth"     "treering"         "trees"             "UCBAdmissions"     "UKDriverDeaths"
[96] "UKgas"           "USAccDeaths"      "USArrests"         "UScitiesD"         "USJudgeRatings"
[101] "USPersonalExpenditure" "uspop"          "VADeaths"         "volcano"           "warpbreaks"
[106] "women"           "WorldPhones"      "WWWusage"
```

```
#Problem 2: Generate a list of datasets within the package ggplot2
```

```
data(package = "ggplot2")
Data sets in package 'ggplot2':
```

diamonds	Prices of over 50,000 round cut
diamonds	
economics	US economic time series
economics_long	US economic time series
faithfuld	2d density estimate of Old Faithful
data	
luv_colours	'colors()' in Luv space
midwest	Midwest demographics
mpg	Fuel economy data from 1999 to 2008
for 38 popular models of cars	
msleep	An updated and expanded version of
the mammals sleep dataset	
presidential	Terms of 12 presidents from
Eisenhower to Trump	
seals	Vector field of seal movements
txhousing	Housing sales in TX

```
#Problem 3 Use head, tail, str, name, glimpse, class on a dataset of Midwest
```

```
> head(midwest)
```

```
# A tibble: 6 × 28
```

	PID	county	state	area	poptotal	popdensity	popwhite	popblack
	<int>	<chr>	<fct>	<dbl>	<int>	<dbl>	<int>	<int>
1	561	ADAMS	IL	0.052	66090	1271.	63917	1702
98	249		124	96.7	2.58		0.148	
2	562	ALEXANDER	IL	0.014	10626	759	7054	3496
19	48		9	66.4	32.9		0.179	
3	563	BOND	IL	0.022	14991	681.	14477	429
35	16		34	96.6	2.86		0.233	
4	564	BOONE	IL	0.017	30806	1812.	29344	127
46	150		1139	95.3	0.412		0.149	
5	565	BROWN	IL	0.018	5836	324.	5264	547
14	5		6	90.2	9.37		0.240	
6	566	BUREAU	IL	0.05	35688	714.	35157	50
65	195		221	98.5	0.140		0.182	

```
# i 14 more variables: percasian <dbl>, percother <dbl>, popadults <int>,
perchsd <dbl>, percollege <dbl>, percprof <dbl>,
# poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
percchildbelowpovert <dbl>, percadultpoverty <dbl>,
# percelderlypoverty <dbl>, inmetro <int>, category <fct>
```

```
> tail(midwest)
```

```
# A tibble: 6 × 28
```

	PID	county	state	area	poptotal	popdensity	popwhite	popblack
	<int>	<chr>	<fct>	<dbl>	<int>	<dbl>	<int>	<int>
1	3047	WASHINGTON	WI	0.025	95328	3813.	94465	125
208	337		193	99.1	0.131		0.218	
2	3048	WAUKESHA	WI	0.034	304715	8962.	298313	1096
672	2699		1935	97.9	0.360		0.221	

```

3 3049 WAUPACA WI 0.045 46104 1025. 45695 22
125 92 170 99.1 0.0477 0.271
4 3050 WAUSHARA WI 0.037 19385 524. 19094 29
70 43 149 98.5 0.150 0.361
5 3051 WINNEBAGO WI 0.035 140320 4009. 136822 697
685 1728 388 97.5 0.497 0.488
6 3052 WOOD WI 0.048 73605 1533. 72157 90
481 722 155 98.0 0.122 0.653
# i 14 more variables: percasian <dbl>, percother <dbl>, popadults <int>,
perchsd <dbl>, percollege <dbl>, percprof <dbl>,
# poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
percchildbelowpovert <dbl>, percadultpoverty <dbl>,
# percelderlypoverty <dbl>, inmetro <int>, category <fct>

> str(midwest)
tibble [437 × 28] (S3: tbl_df/tbl/data.frame)
 $ PID : int [1:437] 561 562 563 564 565 566 567 568 569 570
...
 $ county : chr [1:437] "ADAMS" "ALEXANDER" "BOND" "BOONE" ...
 $ state : Factor w/ 5 levels "IL","IN","MI",...: 1 1 1 1 1 1 1
1 1 1 ...
 $ area : num [1:437] 0.052 0.014 0.022 0.017 0.018 0.05 0.017
0.027 0.024 0.058 ...
 $ poptotal : int [1:437] 66090 10626 14991 30806 5836 35688 5322
16805 13437 173025 ...
 $ popdensity : num [1:437] 1271 759 681 1812 324 ...
 $ popwhite : int [1:437] 63917 7054 14477 29344 5264 35157 5298
16519 13384 146506 ...
 $ popblack : int [1:437] 1702 3496 429 127 547 50 1 111 16 16559
...
 $ popamerindian : int [1:437] 98 19 35 46 14 65 8 30 8 331 ...
 $ popasian : int [1:437] 249 48 16 150 5 195 15 61 23 8033 ...
 $ popother : int [1:437] 124 9 34 1139 6 221 0 84 6 1596 ...
 $ percwhite : num [1:437] 96.7 66.4 96.6 95.3 90.2 ...
 $ percblack : num [1:437] 2.575 32.9 2.862 0.412 9.373 ...
 $ percamerindian : num [1:437] 0.148 0.179 0.233 0.149 0.24 ...
 $ percasian : num [1:437] 0.3768 0.4517 0.1067 0.4869 0.0857 ...
 $ percother : num [1:437] 0.1876 0.0847 0.2268 3.6973 0.1028 ...
 $ popadults : int [1:437] 43298 6724 9669 19272 3979 23444 3583
11323 8825 95971 ...
 $ perchsd : num [1:437] 75.1 59.7 69.3 75.5 68.9 ...
 $ percollege : num [1:437] 19.6 11.2 17 17.3 14.5 ...
 $ percprof : num [1:437] 4.36 2.87 4.49 4.2 3.37 ...
 $ poppovertyknown : int [1:437] 63628 10529 14235 30337 4815 35107 5241
16455 13081 154934 ...
 $ percpovertyknown : num [1:437] 96.3 99.1 95 98.5 82.5 ...
 $ percbelowpoverty : num [1:437] 13.15 32.24 12.07 7.21 13.52 ...
 $ percchildbelowpovert : num [1:437] 18 45.8 14 11.2 13 ...
 $ percadultpoverty : num [1:437] 11.01 27.39 10.85 5.54 11.14 ...
 $ percelderlypoverty : num [1:437] 12.44 25.23 12.7 6.22 19.2 ...
 $ inmetro : int [1:437] 0 0 0 1 0 0 0 0 0 1 ...
 $ category : Factor w/ 16 levels "AAR","AAU","AHR",...: 1 15 1 6 1
1 13 1 1 8 ...

> names(midwest)
 [1] "PID" "county" "state"
"area" "poptotal"

```

[6] "popdensity"	"popwhite"	"popblack"
"popamerindian"	"popasian"	
[11] "popother"	"percwhite"	"percblack"
"percamerindian"	"percasian"	
[16] "percother"	"popadults"	"perchsd"
"percollege"	"percprof"	
[21] "poppovertyknown"	"percpovertyknown"	"percbelowpoverty"
"percchildbelowpovert"	"percadultpoverty"	
[26] "percelderlypoverty"	"inmetro"	"category"

```
> glimpse(midwest)
```

```
Rows: 437
```

```
Columns: 28
```

```
$ PID          <int> 561, 562, 563, 564, 565, 566, 567, 568, 569,
570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582...
$ county       <chr> "ADAMS", "ALEXANDER", "BOND", "BOONE", "BROWN",
"BUREAU", "CALHOUN", "CARROLL", "CASS", "CHAMPAIGN", "CHRIST...
$ state        <fct> IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL,
IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, IL, ...
$ area         <dbl> 0.052, 0.014, 0.022, 0.017, 0.018, 0.050, 0.017,
0.027, 0.024, 0.058, 0.042, 0.030, 0.028, 0.029, 0.030, 0.0...
$ poptotal     <int> 66090, 10626, 14991, 30806, 5836, 35688, 5322,
16805, 13437, 173025, 34418, 15921, 14460, 33944, 51644, 5105...
$ popdensity   <dbl> 1270.9615, 759.0000, 681.4091, 1812.1176,
324.2222, 713.7600, 313.0588, 622.4074, 559.8750, 2983.1897, 819.4...
$ popwhite     <int> 63917, 7054, 14477, 29344, 5264, 35157, 5298,
16519, 13384, 146506, 34176, 15842, 14403, 32688, 50177, 32049...
$ popblack     <int> 1702, 3496, 429, 127, 547, 50, 1, 111, 16,
16559, 82, 10, 4, 1021, 925, 1317147, 63, 5, 2069, 25, 16, 15462,...
$ popamerindian <int> 98, 19, 35, 46, 14, 65, 8, 30, 8, 331, 51, 26,
17, 48, 92, 10289, 34, 6, 123, 37, 19, 962, 24, 8, 45, 40, 14...
$ popasian     <int> 249, 48, 16, 150, 5, 195, 15, 61, 23, 8033, 89,
36, 29, 104, 341, 188565, 48, 26, 1751, 43, 41, 39634, 24, 1...
$ popother     <int> 124, 9, 34, 1139, 6, 221, 0, 84, 6, 1596, 20, 7,
7, 83, 109, 384119, 19, 6, 1021, 24, 108, 10703, 10, 6, 29,...
$ percwhite    <dbl> 96.71206, 66.38434, 96.57128, 95.25417,
90.19877, 98.51210, 99.54904, 98.29813, 99.60557, 84.67331, 99.29688...
$ percblack    <dbl> 2.57527614, 32.90043290, 2.86171703, 0.41225735,
9.37285812, 0.14010312, 0.01878993, 0.66051770, 0.11907420,...
$ percamerindian <dbl> 0.14828264, 0.17880670, 0.23347342, 0.14932156,
0.23989034, 0.18213405, 0.15031943, 0.17851830, 0.05953710, ...
$ percasian    <dbl> 0.37675897, 0.45172219, 0.10673071, 0.48691813,
0.08567512, 0.54640215, 0.28184893, 0.36298721, 0.17116916, ...
$ percother    <dbl> 0.18762294, 0.08469791, 0.22680275, 3.69733169,
0.10281014, 0.61925577, 0.00000000, 0.49985123, 0.04465282, ...
$ popadults    <int> 43298, 6724, 9669, 19272, 3979, 23444, 3583,
11323, 8825, 95971, 22945, 10734, 9647, 21563, 29136, 3291995, ...
$ perchsd      <dbl> 75.10740, 59.72635, 69.33499, 75.47219,
68.86152, 76.62941, 62.82445, 75.95160, 72.27195, 87.49935, 73.07474...
$ percollege   <dbl> 19.63139, 11.24331, 17.03382, 17.27895,
14.47600, 18.90462, 11.91739, 16.19712, 14.10765, 41.29581, 13.56723...
$ percprof     <dbl> 4.355859, 2.870315, 4.488572, 4.197800,
3.367680, 3.275891, 3.209601, 3.055727, 3.206799, 17.757448, 3.08999...
$ poppovertyknown <int> 63628, 10529, 14235, 30337, 4815, 35107, 5241,
16455, 13081, 154934, 33788, 15615, 14248, 32190, 45693, 5023...
$ percpovertyknown <dbl> 96.27478, 99.08714, 94.95697, 98.47757,
82.50514, 98.37200, 98.47802, 97.91729, 97.35060, 89.54429, 98.16956...
```

```

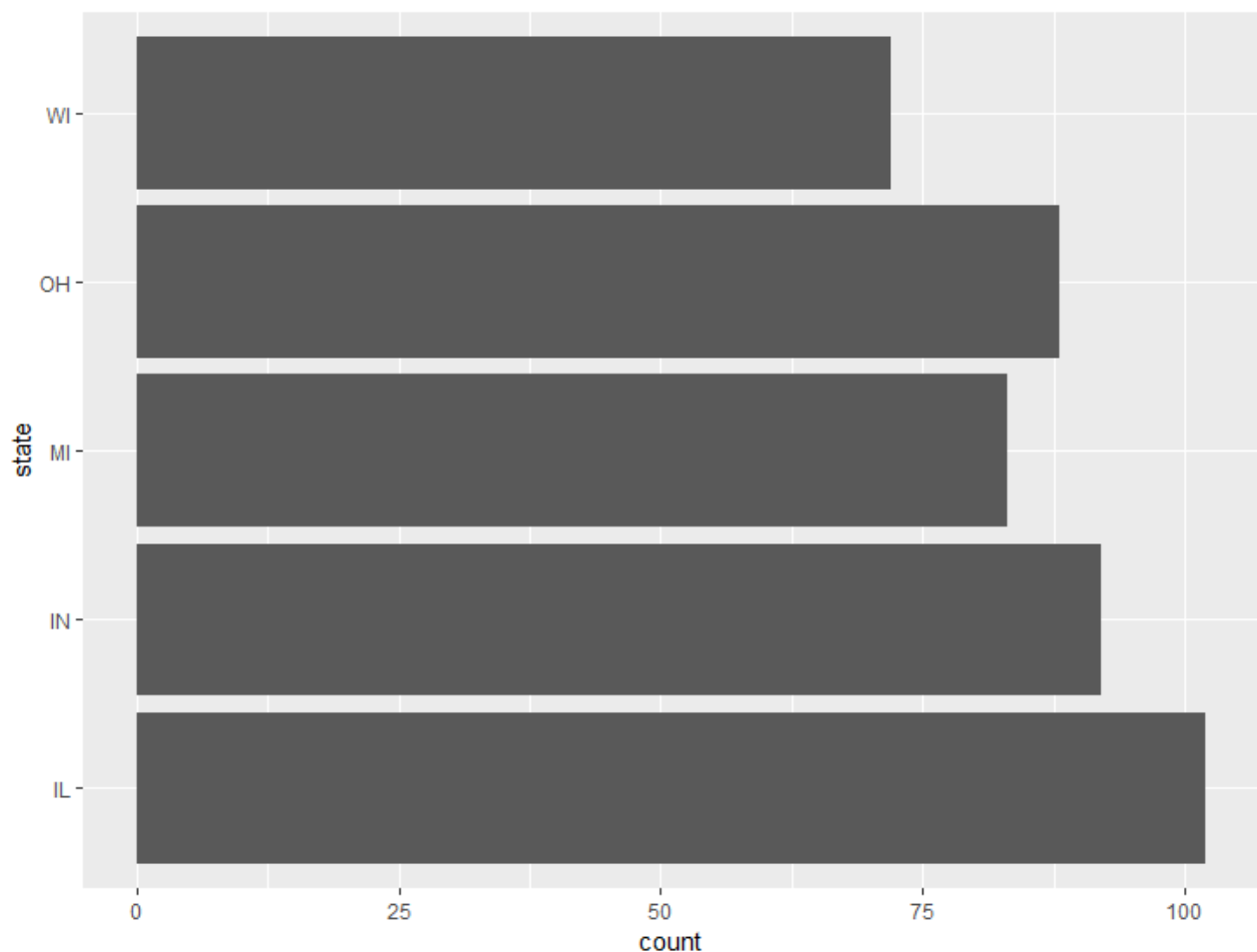
$ percbelowpoverty      <dbl> 13.151443, 32.244278, 12.068844, 7.209019,
13.520249, 10.399635, 15.149781, 11.710726, 13.875086, 15.572437,...
$ percchildbelowpovert <dbl> 18.011717, 45.826514, 14.036061, 11.179536,
13.022889, 14.158819, 13.787761, 17.225462, 17.994784, 14.132234...
$ percadultpoverty      <dbl> 11.009776, 27.385647, 10.852090, 5.536013,
11.143211, 8.179287, 12.932331, 10.027037, 11.914343, 17.562728, ...
$ percelderlypoverty    <dbl> 12.443812, 25.228976, 12.697410, 6.217047,
19.200000, 11.008586, 21.085271, 9.525052, 13.660180, 8.105017, 1...
$ inmetro               <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 1,
0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ...
$ category              <fct> AAR, LHR, AAR, ALU, AAR, AAR, LAR, AAR, AAR,
HAU, AAR, AAR, LAR, LAU, AAR, AAU, AAR, AAR, HAU, AAR, AAR, HLU...

> class(midwest)
[1] "tbl_df"      "tbl"        "data.frame"

```

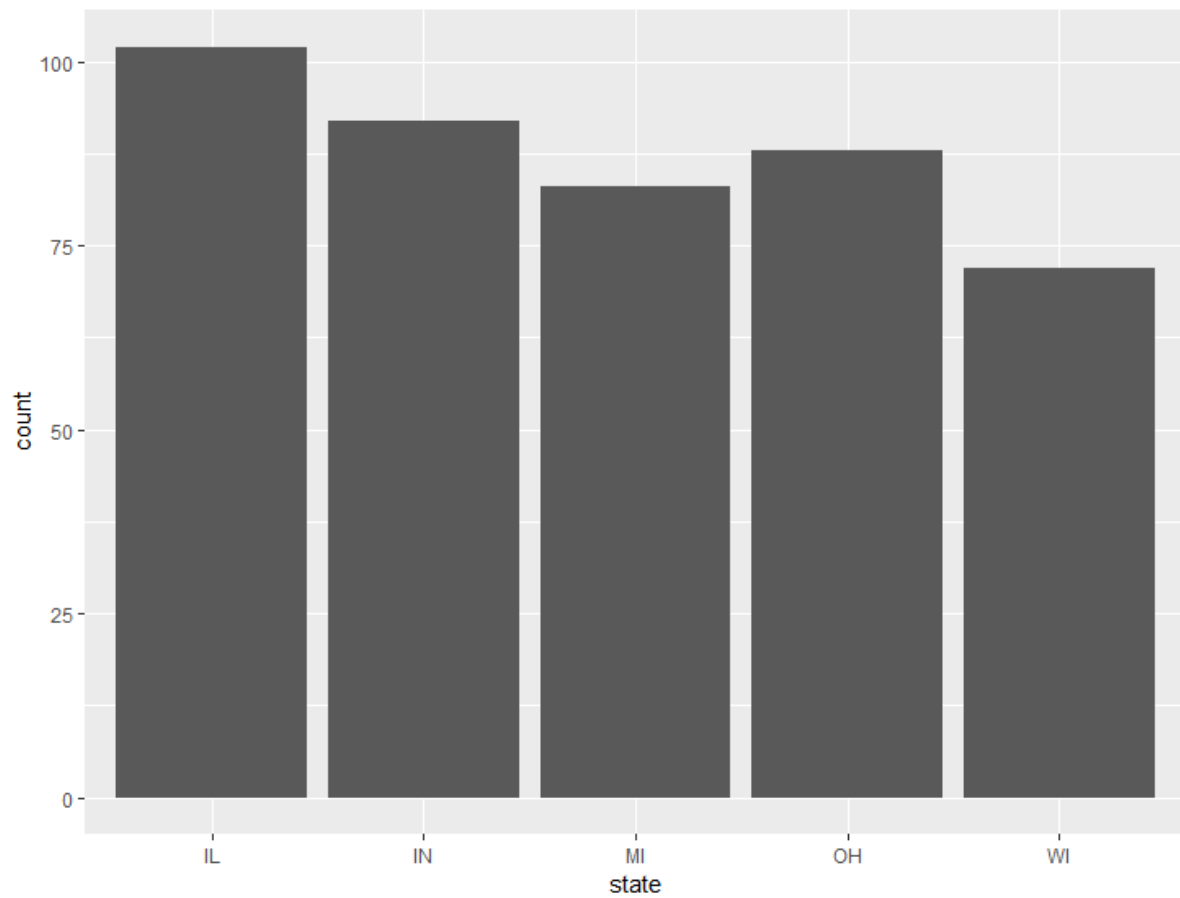
#Problem 4 Create a horizontal bar chart on data in midwest

```
> ggplot(midwest)+geom_bar(aes(y=state))
```

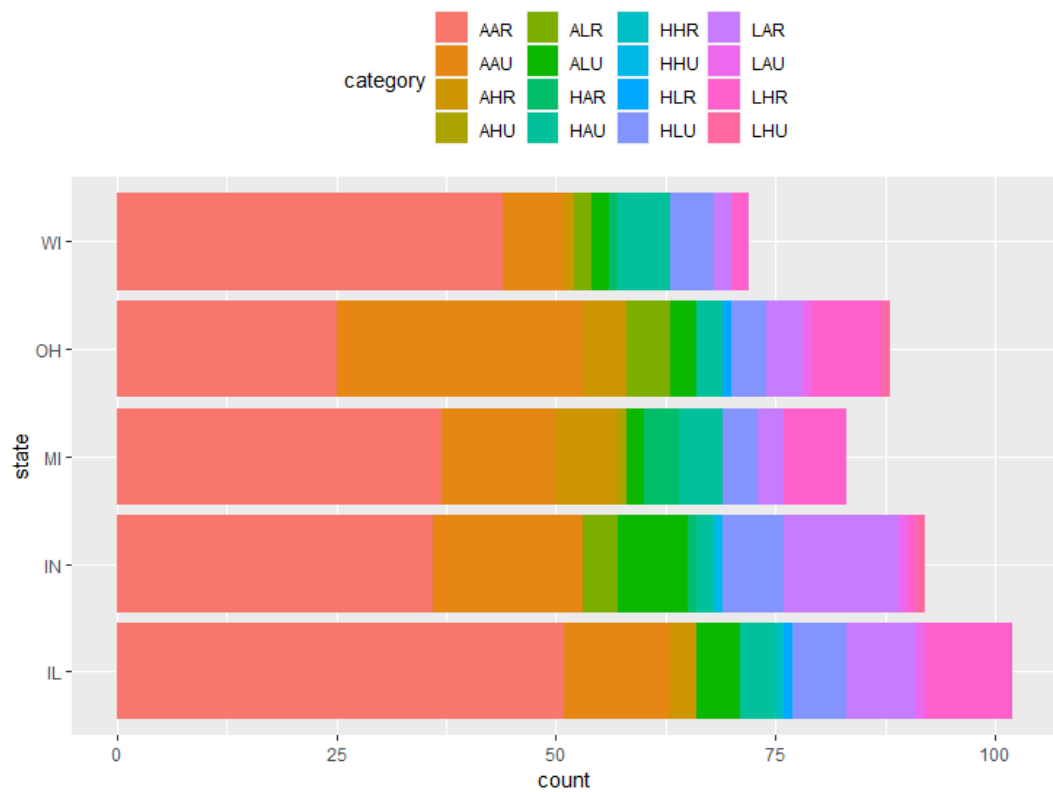


#Problem 5 Create a vertical bar chart on data in midwest

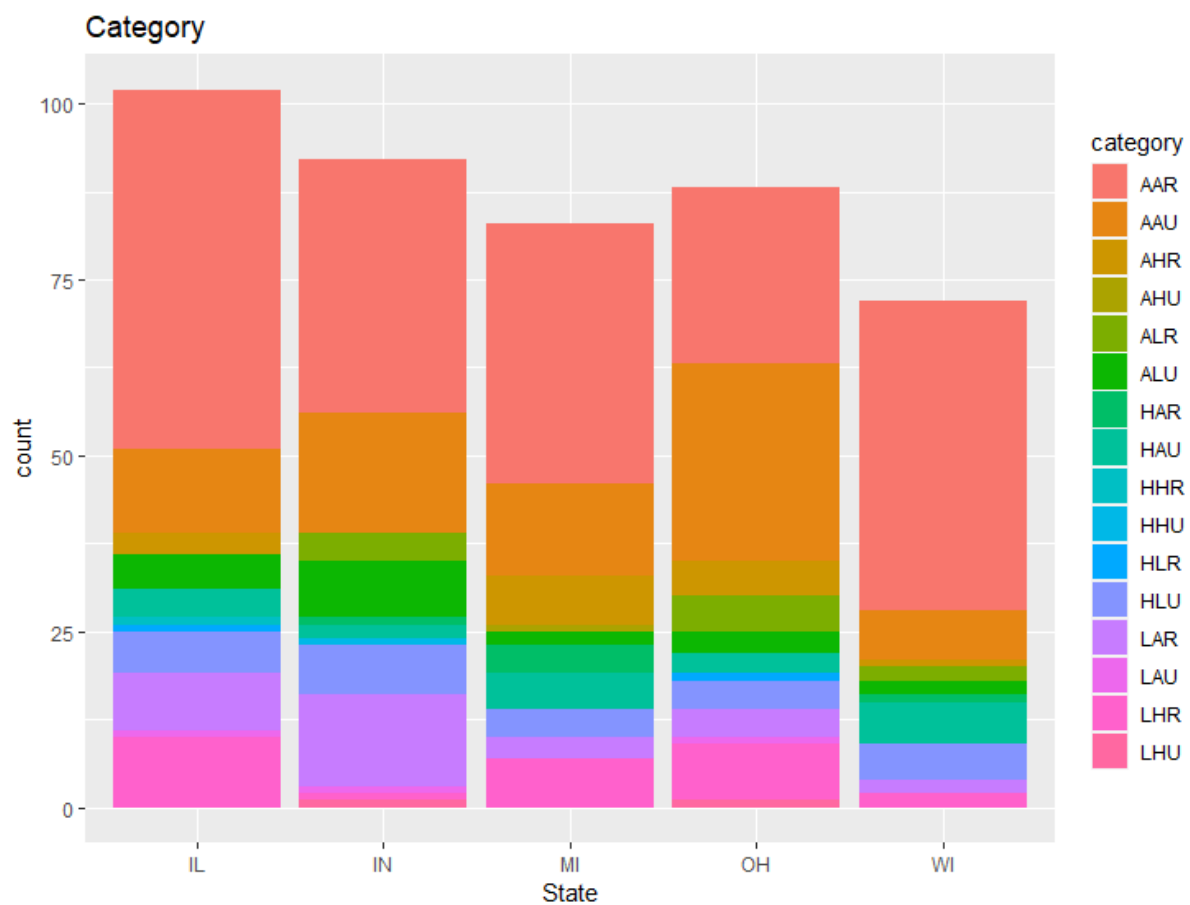
```
> ggplot(midwest)+geom_bar(aes(y=state))+  
  labs()+  
  coord_flip()
```



```
ggplot(midwest, aes(y=state, fill=category))+
  geom_bar(position=position_stack(reverse=TRUE))+
  theme(legend.position="top")
```



```
midwest$state<-factor(midwest$state)
midwest$category<-factor(midwest$category)
ggplot(data=midwest, aes(x=state, fill=category))+
  geom_bar()+
  xlab("State")+
  ggtitle("Category")
```



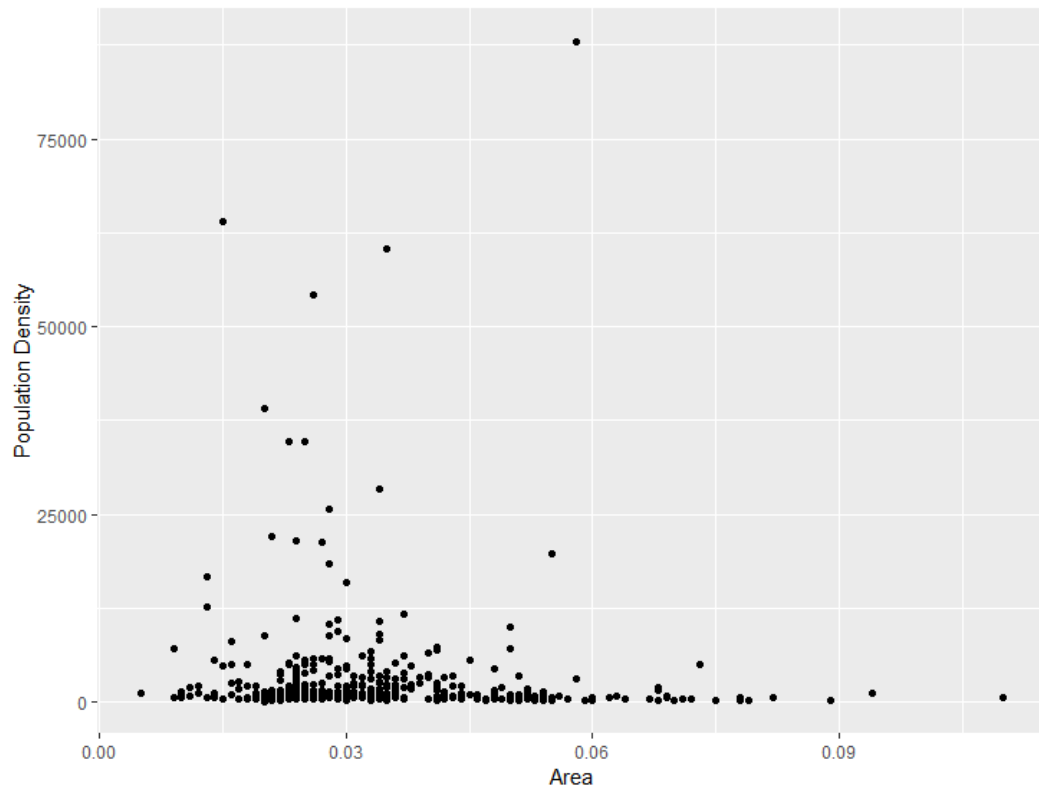
#Problem 8 Create a scatterplot on data in midwest

```
ggplot(midwest, aes(x=area, y=popdensity))+
```

```
  geom_point()+
```

```
  xlab("Area")+
```

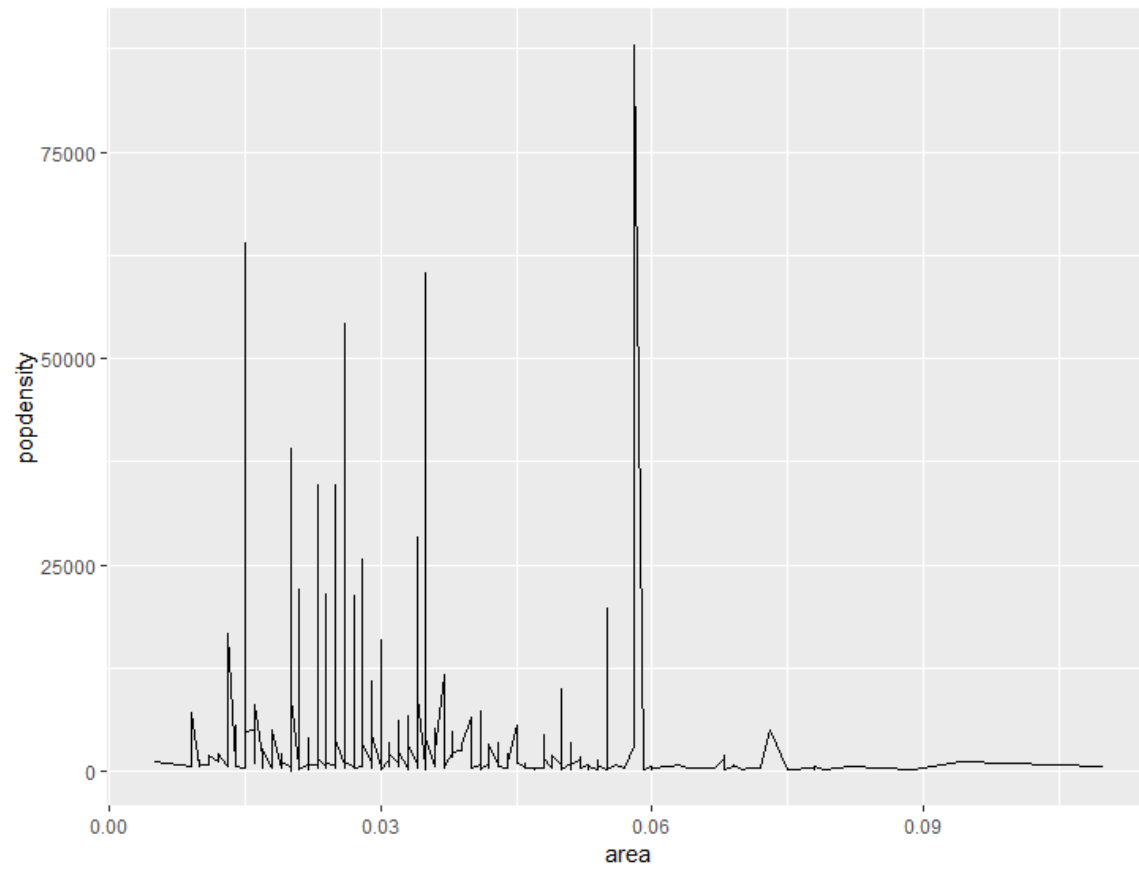
```
  ylab("Population Density")
```



#Problem 9 Create a line chart on data in midwest

```
ggplot(midwest, aes(x=area, y=popdensity))+
```

```
geom_line()
```



#Problem 10 Create a histogram on data in midwest

```
ggplot(midwest, aes(x=popdensity))+  
  geom_histogram(fill="purple",alpha=0.7, binwidth=10000)+  
  xlab("pop density")+  
  ylab("Frequency")+  
  theme_bw()  
  
geom_histogram(fill="purple", alpha=0.7, binwidth=10000) +  
xlab("pop density") +  
ylab("Frequency") +  
theme_bw()
```

